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VIBRATION FAULT DIAGNOSTICS IN MOTORS

School of Engineering | Mechatronic Department | In Development

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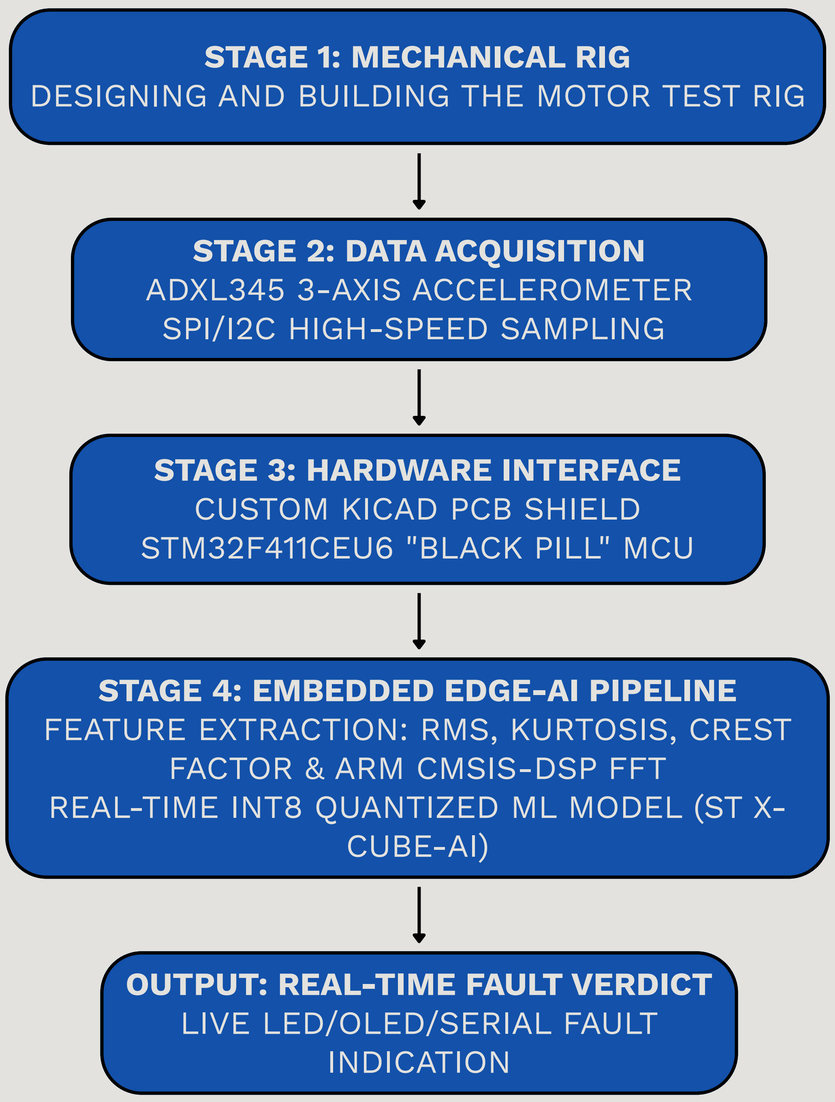
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BACKGROUND

Electric motors drive critical industrial equipment like pumps and fans, emitting unique vibration signatures long before catastrophic mechanical failure occurs.

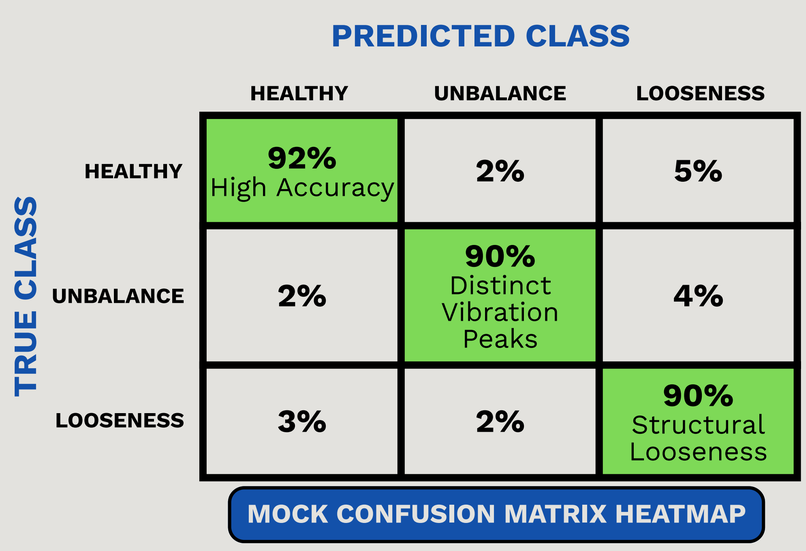
Question: How can a low-cost STM32 microcontroller use Edge AI to classify motor faults in real time without cloud infrastructure or manual inspections?

METHODOLOGY



RESULTS

EXPECTED RESULTS



- **High classification accuracy (>=90%)** across targeted motor states (Healthy, Unbalance, Structural Looseness) evaluated on the test rig.
- **Low Latency (< 100 ms):** Rapid on-chip feature extraction and INT8 quantized execution.
- **Minimal Memory:** Operates well within STM32F411 limits (512 KB Flash 128 KB SRAM).
- **Fully Standalone:** 100% offline, edge-level fault detection with zero cloud or PC dependency.

CONCLUSION

- **Effective Edge Monitoring:** Demonstrates a low-cost, standalone MCU system for real-time motor fault detection without cloud reliance.
- **Optimized Pipeline:** Leverages direct adhesive sensing, time/frequency feature extraction, and INT8 model quantization for high accuracy (>= 90%).
- **Practical Industrial Alternative:** Eliminates high network latency and manual inspection bottlenecks through localized, sub-100 ms processing.

REFERENCES

[1] R. B. RANDALL, VIBRATION-BASED CONDITION MONITORING: INDUSTRIAL, AEROSPACE AND AUTOMOTIVE APPLICATIONS, 1ST ED. CHICHESTER, U.K.: JOHN WILEY & SONS, 2011.

— CORE THEORETICAL FOUNDATION FOR TIME-DOMAIN (RMS, KURTOSIS, CREST FACTOR) AND FREQUENCY-DOMAIN (FFT) SIGNAL PROCESSING.

Project Repository